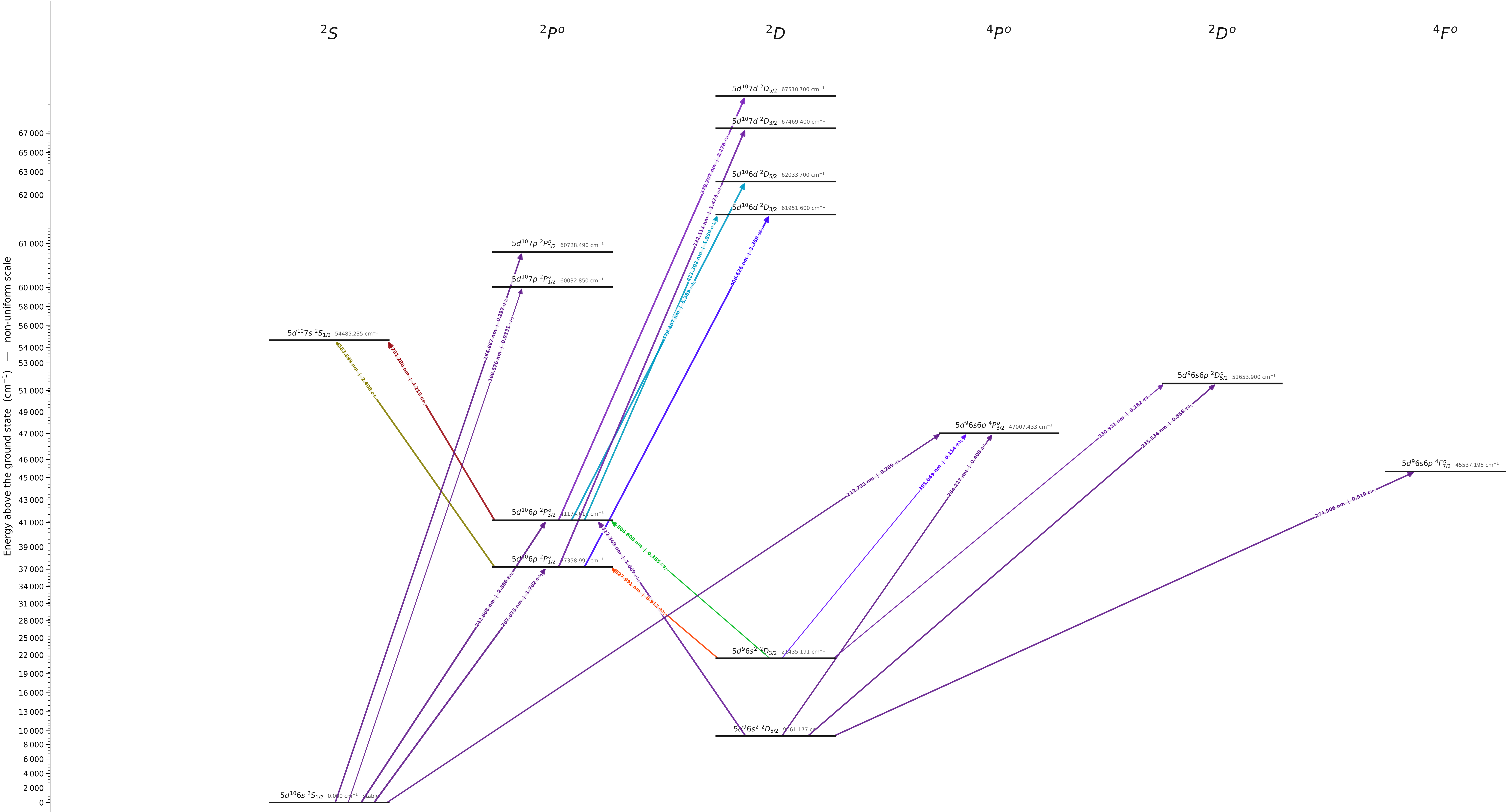
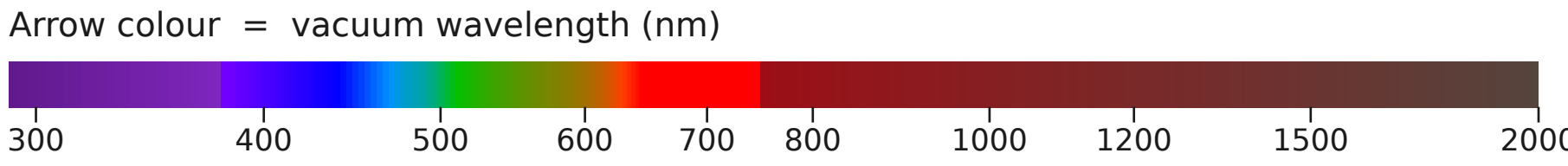


Gold

Au (Au I). Strongest classified transitions below 2 μm (NIST): 15 levels, 20 lines.

Au ⁺ ionisation limit 74409.11 cm⁻¹ = 9.22555 eV (134.392 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(J||e r||J')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(J e r J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d ¹⁰ 6s ² S _{1/2} – 5d ¹⁰ 7p ² P ^o _{3/2}	164.6674	–	1820.5943	0.297	1.000e+07	–	–	–
5d ¹⁰ 6s ² S _{1/2} – 5d ⁹ 6s6p ⁴ P ^o _{3/2}	212.7323	212.6651	1409.2474	0.269	3.800e+06	–	–	–
5d ⁹ 6s ² ² D _{3/2} – 5d ⁹ 6s6p ² D ^o _{3/2}	235.3344	235.2624	1273.8998	0.556	8.000e+06	–	–	–
5d ¹⁰ 6s ² S _{1/2} – 5d ¹⁰ 6p ² P ^o _{3/2}	242.8681	242.7944	1234.3838	2.366	1.980e+08	–	–	–
5d ⁹ 6s ² ² D _{3/2} – 5d ⁹ 6s6p ⁴ P ^o _{3/2}	264.2269	264.1482	1134.6022	0.400	4.400e+06	–	–	–
5d ¹⁰ 6s ² S _{1/2} – 5d ¹⁰ 6p ² P ^o _{1/2}	267.6732	267.5937	1119.9944	1.762	1.640e+08	–	–	–
5d ⁹ 6s ² ² D _{3/2} – 5d ⁹ 6s6p ⁴ F ^o _{7/2}	274.9064	274.8251	1090.5256	0.919	1.030e+07	–	–	–
5d ⁹ 6s ² ² D _{3/2} – 5d ¹⁰ 6p ² P ^o _{3/2}	312.3688	312.2783	959.7387	1.069	1.900e+07	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 7d ² D _{3/2}	332.1111	332.0155	902.6874	1.473	3.000e+07	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 7d ² D _{5/2}	379.7071	379.5993	789.5360	2.278	3.200e+07	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 6d ² D _{3/2}	406.6262	406.5114	737.2679	3.359	8.500e+07	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 6d ² D _{5/2}	479.4074	479.2734	625.3397	5.389	8.900e+07	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 6d ² D _{3/2}	481.3017	481.1673	622.8784	1.859	1.570e+07	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 7s ² S _{1/2}	583.8892	583.7374	513.4319	2.408	2.950e+07	–	–	–
5d ⁹ 6s ² ² D _{3/2} – 5d ¹⁰ 6p ² P ^o _{1/2}	627.9908	627.8172	477.3835	0.912	3.400e+06	–	–	–
5d ¹⁰ 6p ² P ^o _{3/2} – 5d ¹⁰ 7s ² S _{1/2}	751.2797	751.0729	399.0424	4.213	4.240e+07	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).